

Lecture 4

MOVING AVERAGE MODELS

Review: Autoregressive Model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Autoregressive model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

Prediction depends on the previous value of Z_{t-1} .

Review: Autoregressive Model

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$$Z_t - E(Z_t \mid \mathcal{Z}_{t-1}) = Z_t - \phi_1 Z_{t-1}$$

Review: Autoregressive Model

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Autoregressive model for deviations from trend:

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$$Z_t - E(Z_t \mid \mathcal{Z}_{t-1}) = Z_t - \phi_1 Z_{t-1}$$

One-step-ahead
prediction error $\rightarrow U_t = Z_t - \phi_1 Z_{t-1}$

Review: Autoregressive Model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Autoregressive model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \phi_1 Z_{t-1}$$

$$\textcolor{red}{U}_t = Z_t - \phi_1 Z_{t-1}$$

$$Z_t = \phi_1 Z_{t-1} + \textcolor{red}{U}_t$$

Review: Autoregressive Model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Autoregressive model for deviations from trend:

$$Z_t = \phi_1 Z_{t-1} + U_t$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \theta_1 (Z_{t-1} - E(Z_{t-1} \mid \mathcal{Z}_{t-2}))$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \theta_1 (Z_{t-1} - E(Z_{t-1} \mid \mathcal{Z}_{t-2}))$$

Prediction depends on the previous *prediction error*.

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$\begin{aligned} E(Z_t \mid \mathcal{Z}_{t-1}) &= \theta_1 (Z_{t-1} - E(Z_{t-1} \mid \mathcal{Z}_{t-2})) \\ &= \theta_1 U_{t-1} \end{aligned}$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \theta_1 U_{t-1}$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \theta_1 U_{t-1}$$

$$Z_t - E(Z_t \mid \mathcal{Z}_{t-1}) = Z_t - \theta_1 U_{t-1}$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$E(Z_t \mid \mathcal{Z}_{t-1}) = \theta_1 U_{t-1}$$

$$Z_t - E(Z_t \mid \mathcal{Z}_{t-1}) = Z_t - \theta_1 U_{t-1}$$

$$U_t = Z_t - \theta_1 U_{t-1}$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$U_t = Z_t - \theta_1 U_{t-1}$$

$$Z_t = U_t + \theta_1 U_{t-1}$$

Moving average model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

Moving average model for deviations from trend:

$$Z_t = U_t + \theta_1 U_{t-1}$$

AR(p) model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

AR(p) model for deviations from trend:

$$Z_t = \phi_1 Z_{t-1} + \dots + \phi_p Z_{t-p} + U_t$$

MA(q) model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

MA(q) model for deviations from trend:

$$Z_t = U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q}$$

ARMA(p,q) model

Deterministic trend model:

$$Y_t = X_t' \beta + Z_t$$

ARMA(p,q) model for deviations from trend:

$$\begin{aligned} Z_t = & \phi_1 Z_{t-1} + \dots + \phi_p Z_{t-p} \\ & + U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q} \end{aligned}$$

ARMA(p,q) model selection

Same *practical* approach as AR(p) selection:

- check residual autocorrelation tests
- choose model with smallest AICc

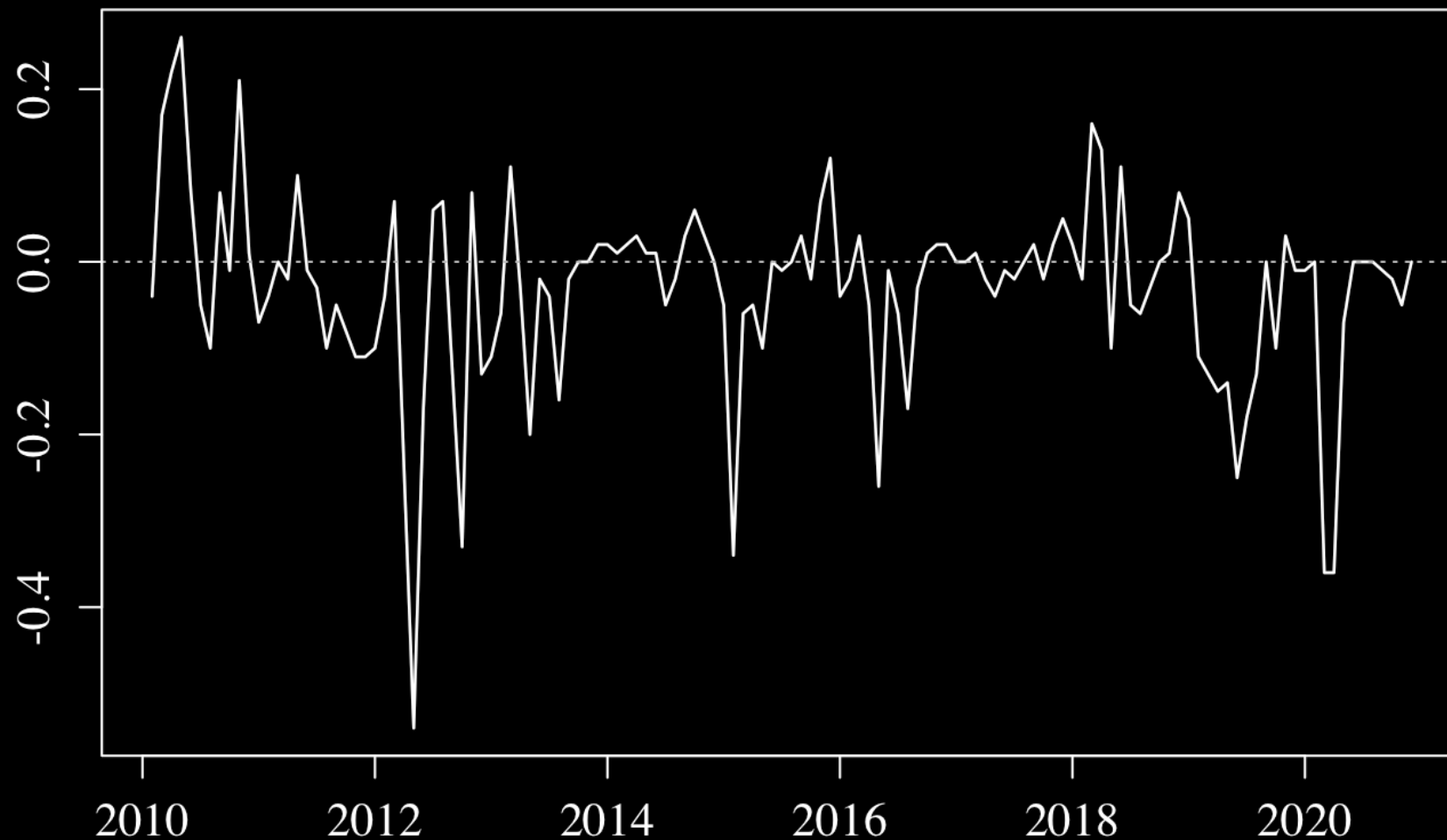
Also *in theory* we can look at autocorrelations of the time series.

First difference of interest rates

(3 month Bank Accepted Bills)

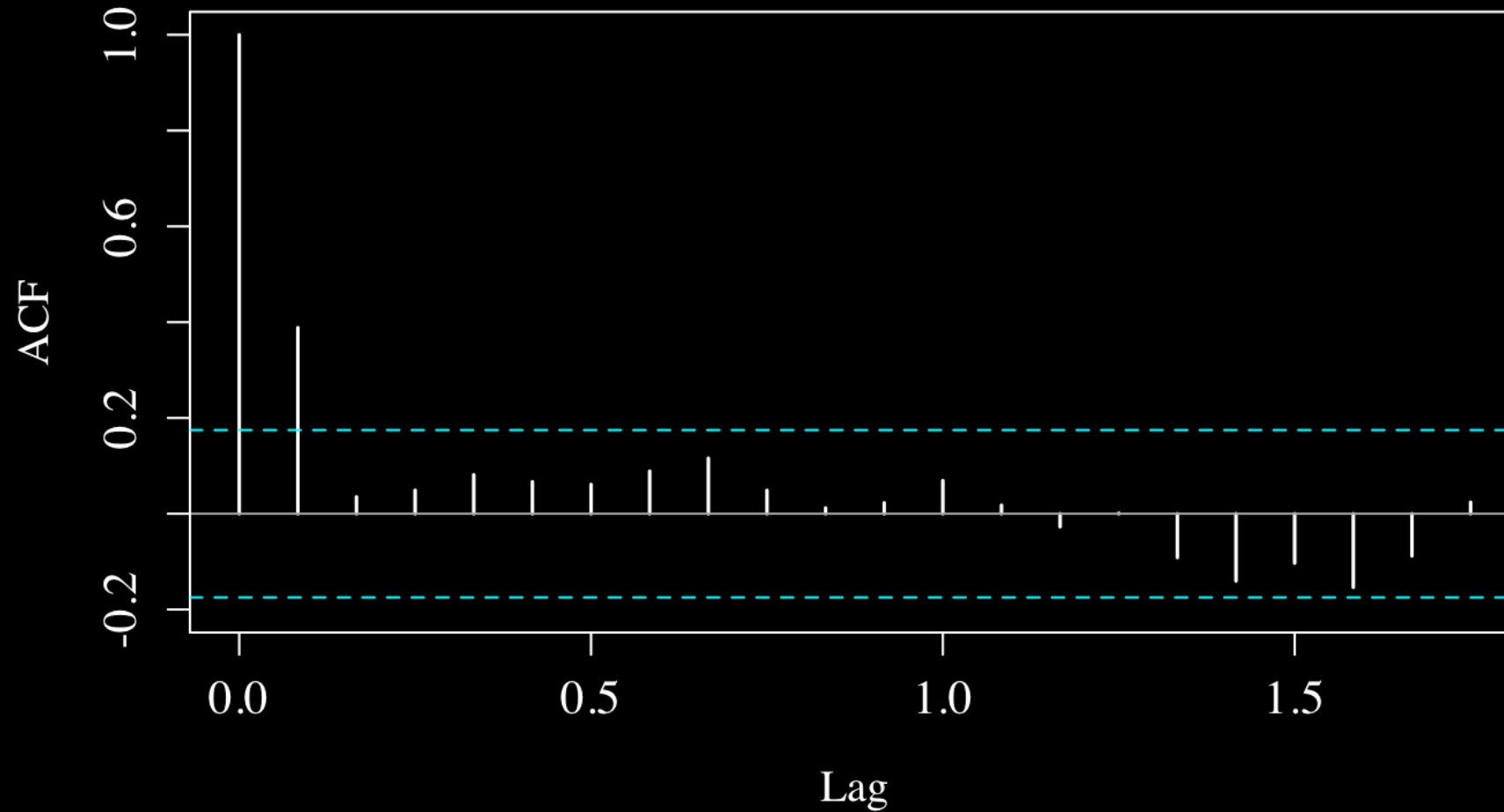
```
1 dt <- read.csv("BAB3mth.csv")
2 Y <- ts(dt$BAB3,
3         start=c(2010,1),
4         end=c(2025,6),
5         frequency=12)
6 Y <- window(Y, end=c(2020,12))
7 DY <- diff(Y)
```

First difference of interest rates (3 month Bank Accepted Bills)



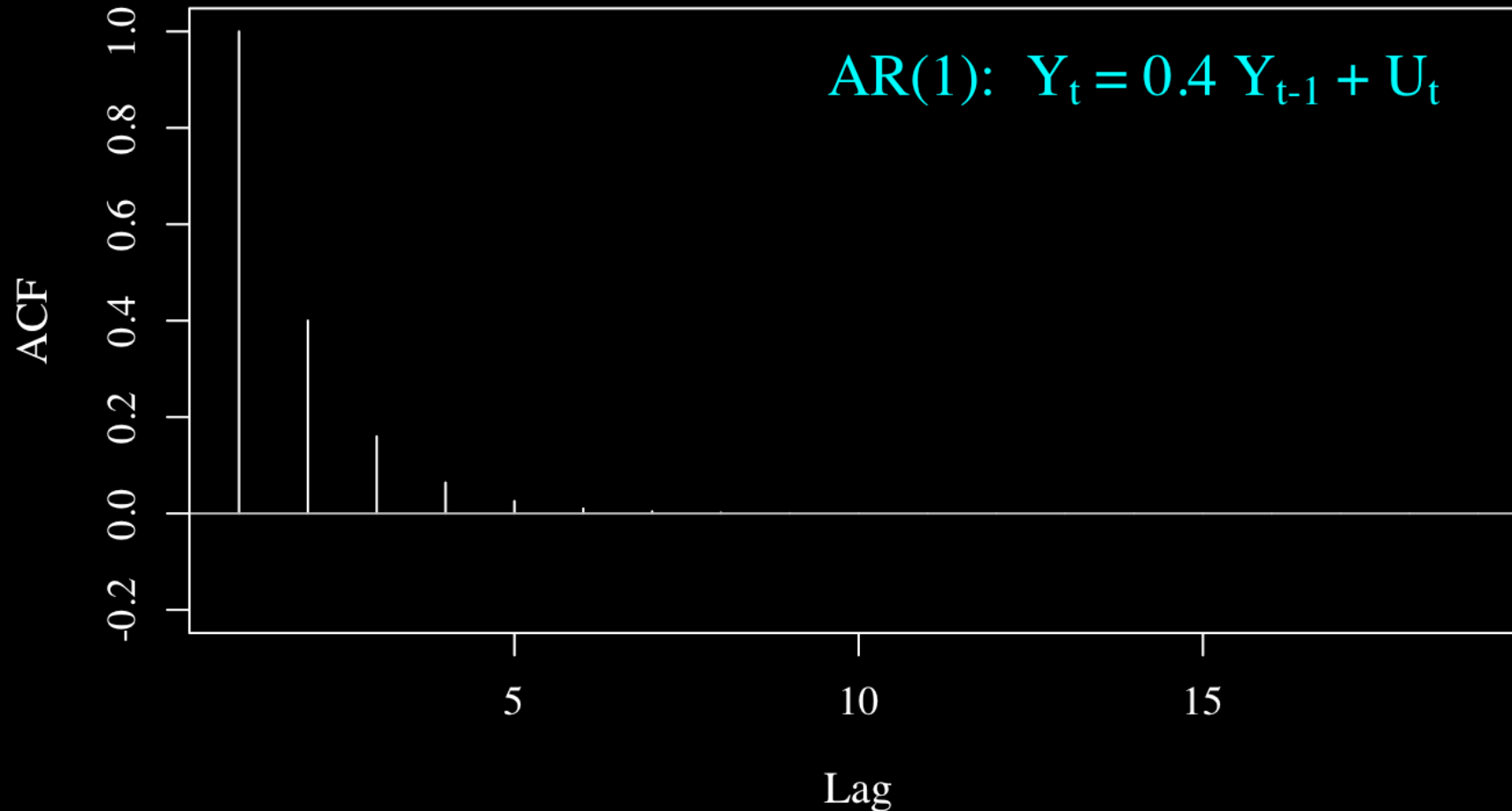
First difference of interest rates

```
1 acf(DY)
```



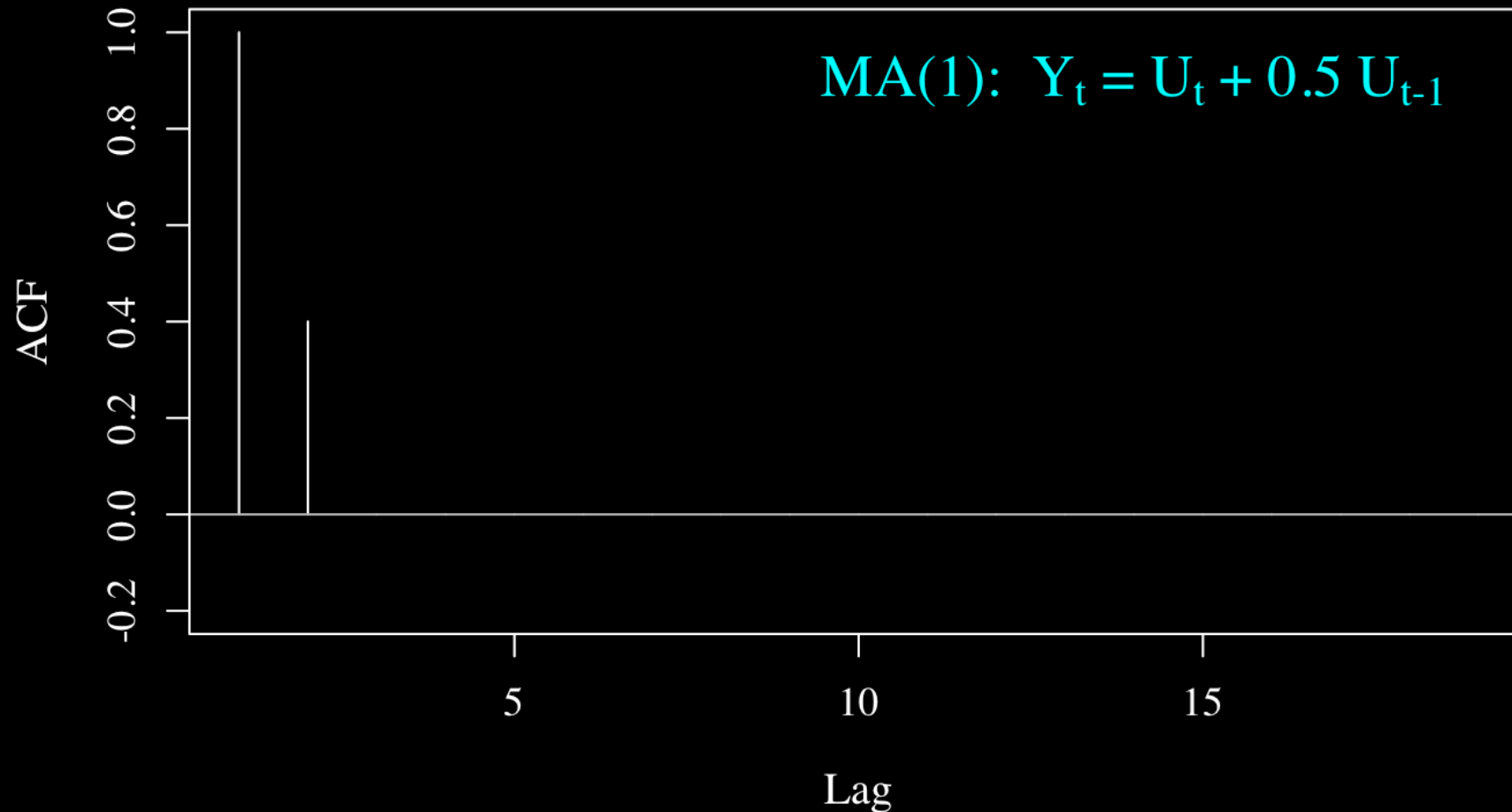
Theoretical autocorrelation function

```
1 acf_AR1 <- ARMAacf(ar=0.4, lag.max=18)
```



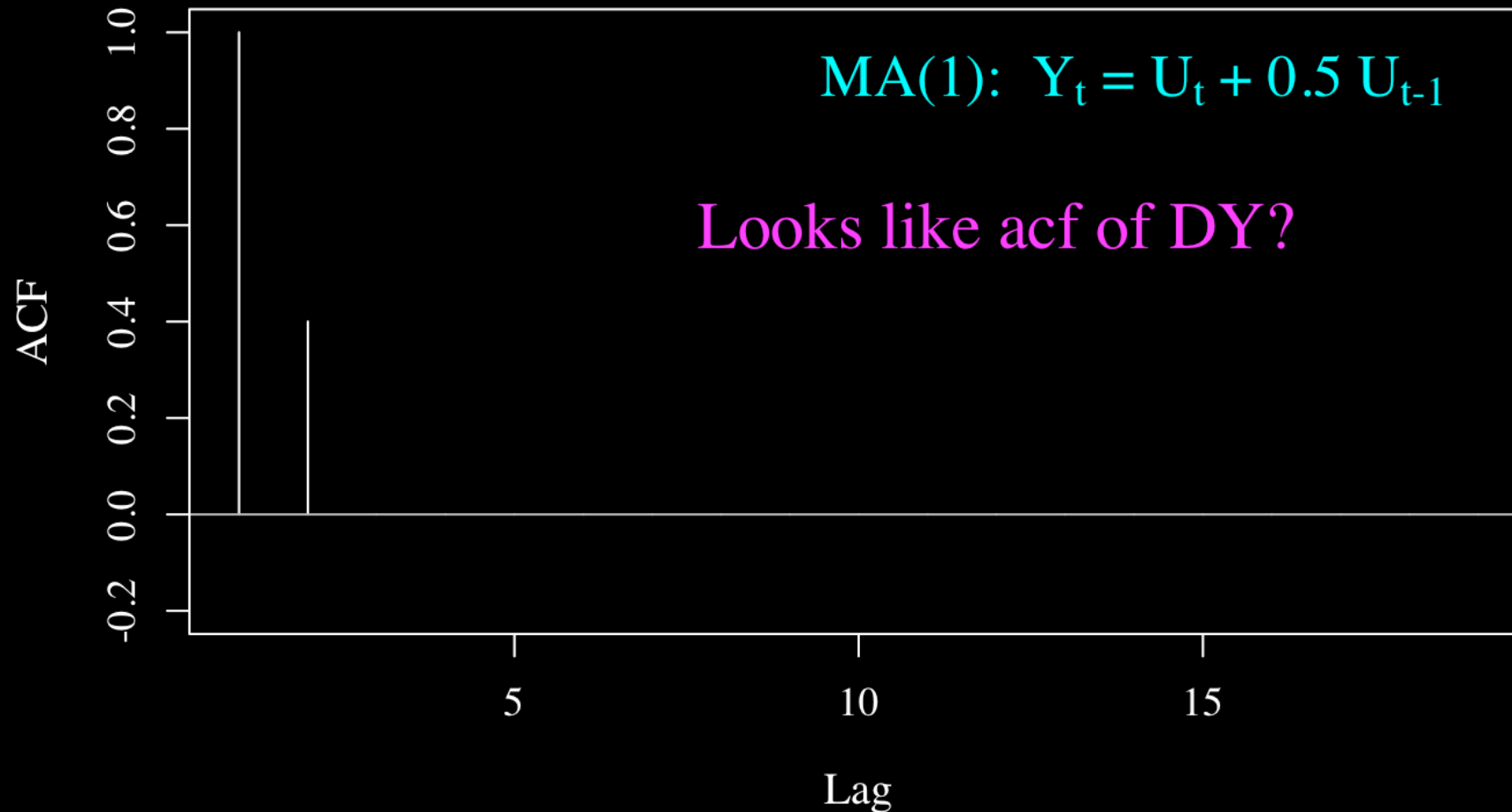
Theoretical autocorrelation function

```
1 acf_MA1 <- ARMAacf(ma=0.5, lag.max=18)
```



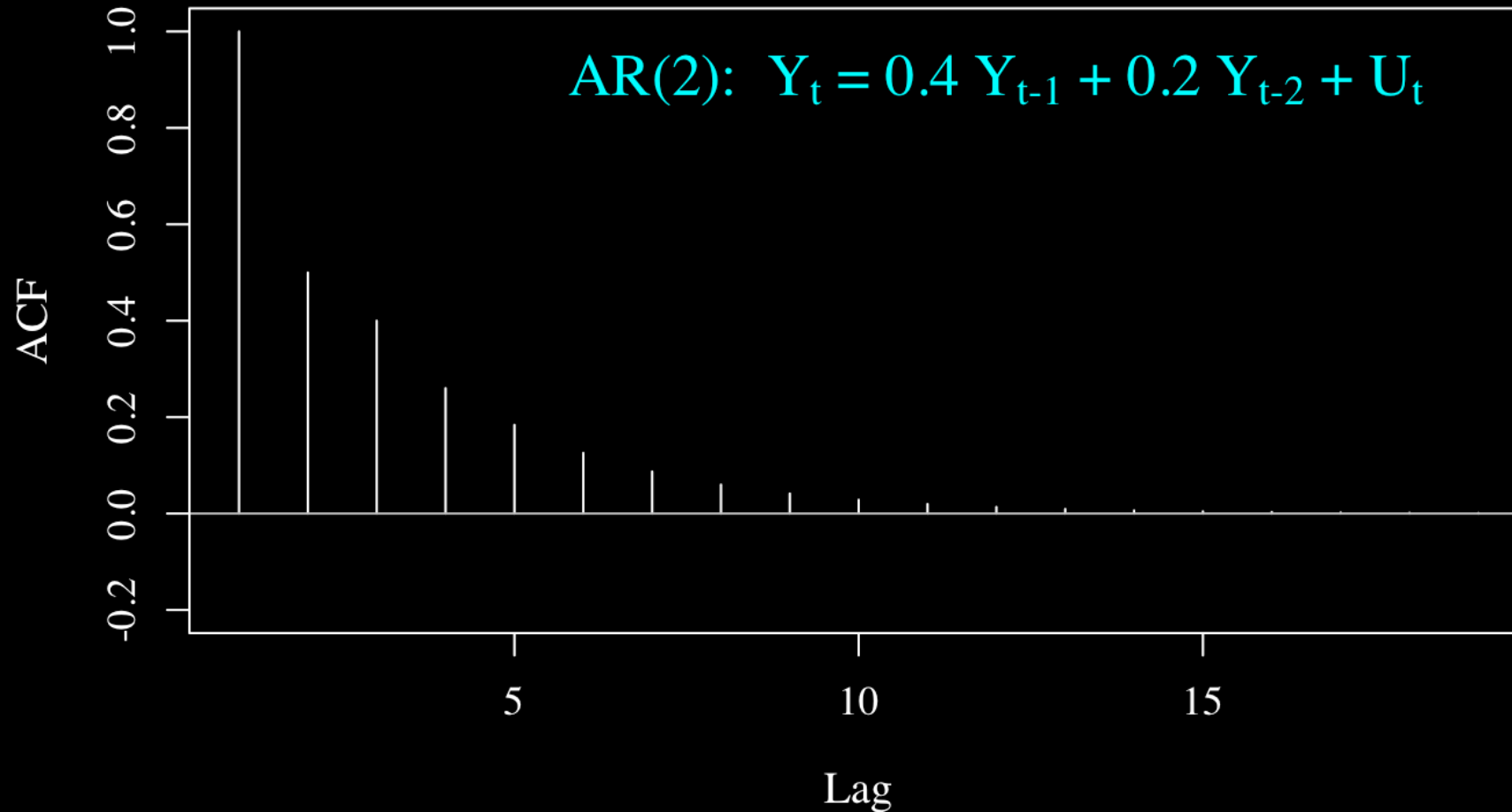
Theoretical autocorrelation function

```
1 acf_MA1 <- ARMAacf(ma=0.5, lag.max=18)
```



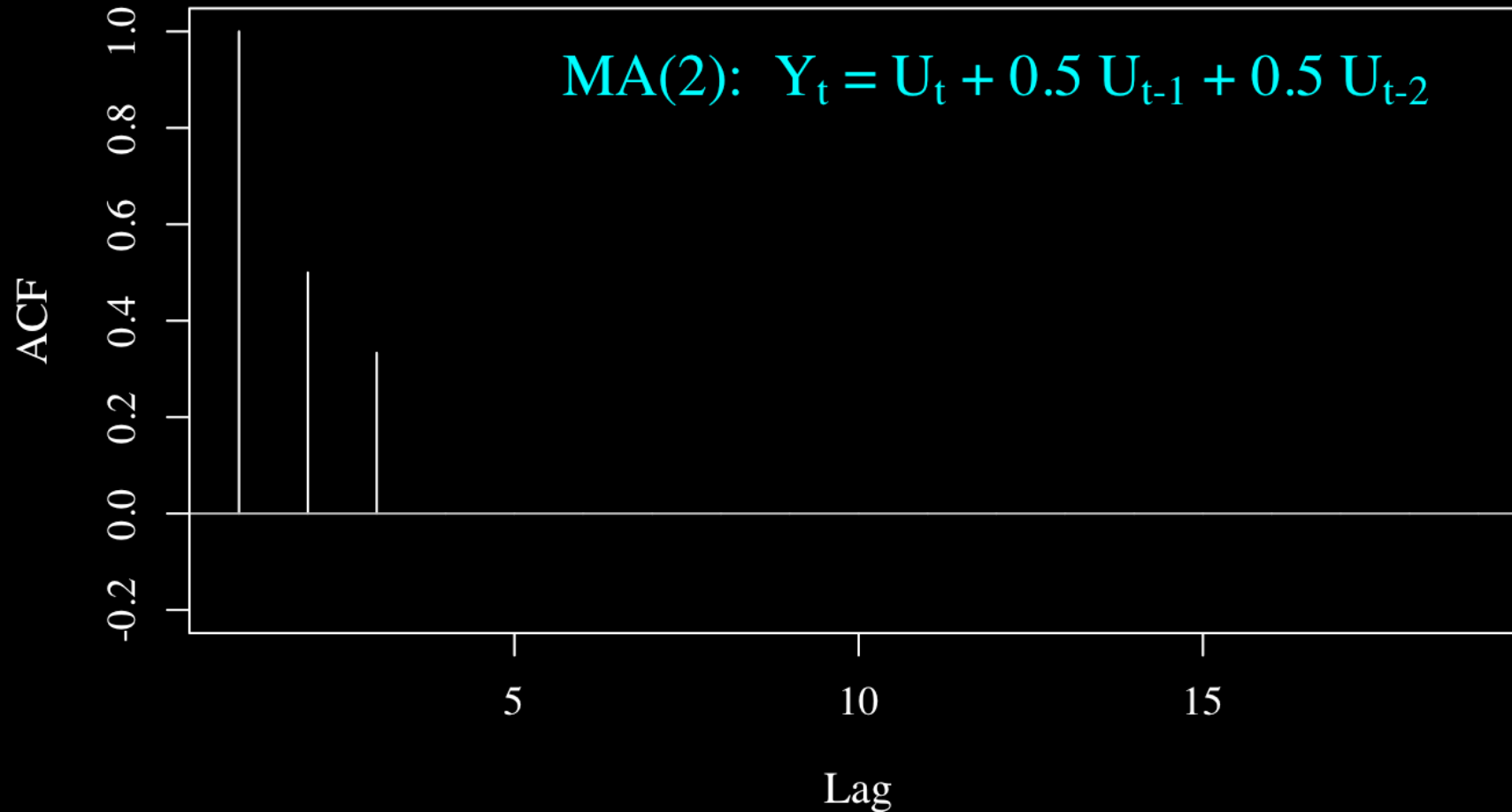
Some other theoretical acf's

```
1 acf_AR2 <- ARMAacf(ar=c(0.4,0.2), lag.max=1
```



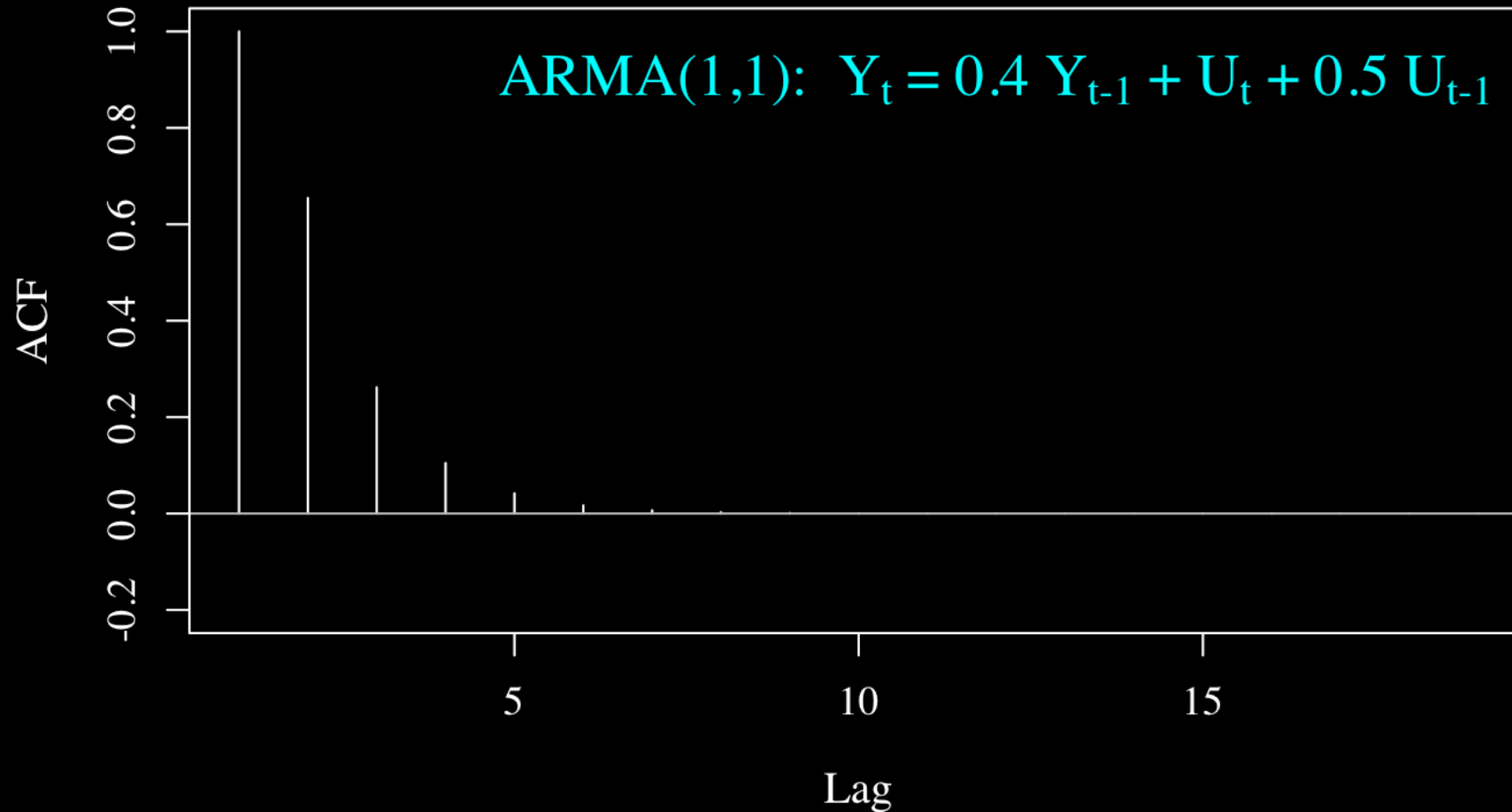
Some other theoretical acf's

```
1 acf_MA2 <- ARMAacf(ma=c(0.5,0.5), lag.max=1
```



Some other theoretical acf's

```
1 acf_ARMA11 <- ARMAacf(ar=0.4, ma=0.5, lag.m
```



Partial autocorrelation function

Recall autocorrelation at lag k :

correlation between Y_t and Y_{t-k} for any $k = 1, 2, 3, \dots$

Partial autocorrelation at lag k :

correlation between Y_t and Y_{t-k} *after controlling for intermediate lags* $Y_{t-1}, \dots, Y_{t-k+1}$.

Partial autocorrelation function

Partial autocorrelation at lag k :

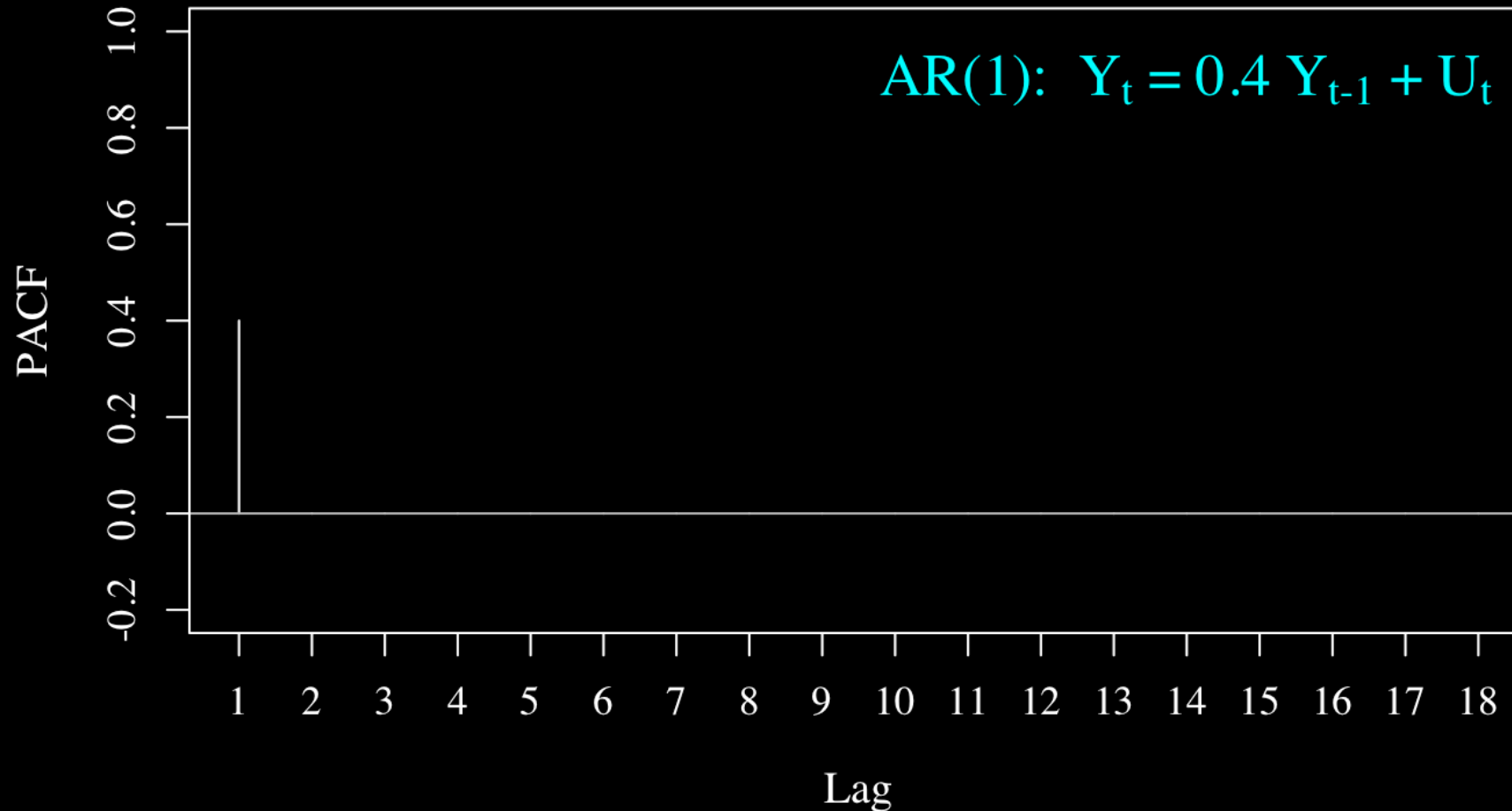
correlation between Y_t and Y_{t-k} *after controlling for intermediate lags* $Y_{t-1}, \dots, Y_{t-k+1}$.

i.e. correlation between $Y_{t|k}^*$ and $Y_{t-k|k}^*$ where

$Y_{t|k}^*$ and $Y_{t-k|k}^*$ are residuals from regressions of Y_t and Y_{t-k} on $Y_{t-1}, \dots, Y_{t-k+1}$.

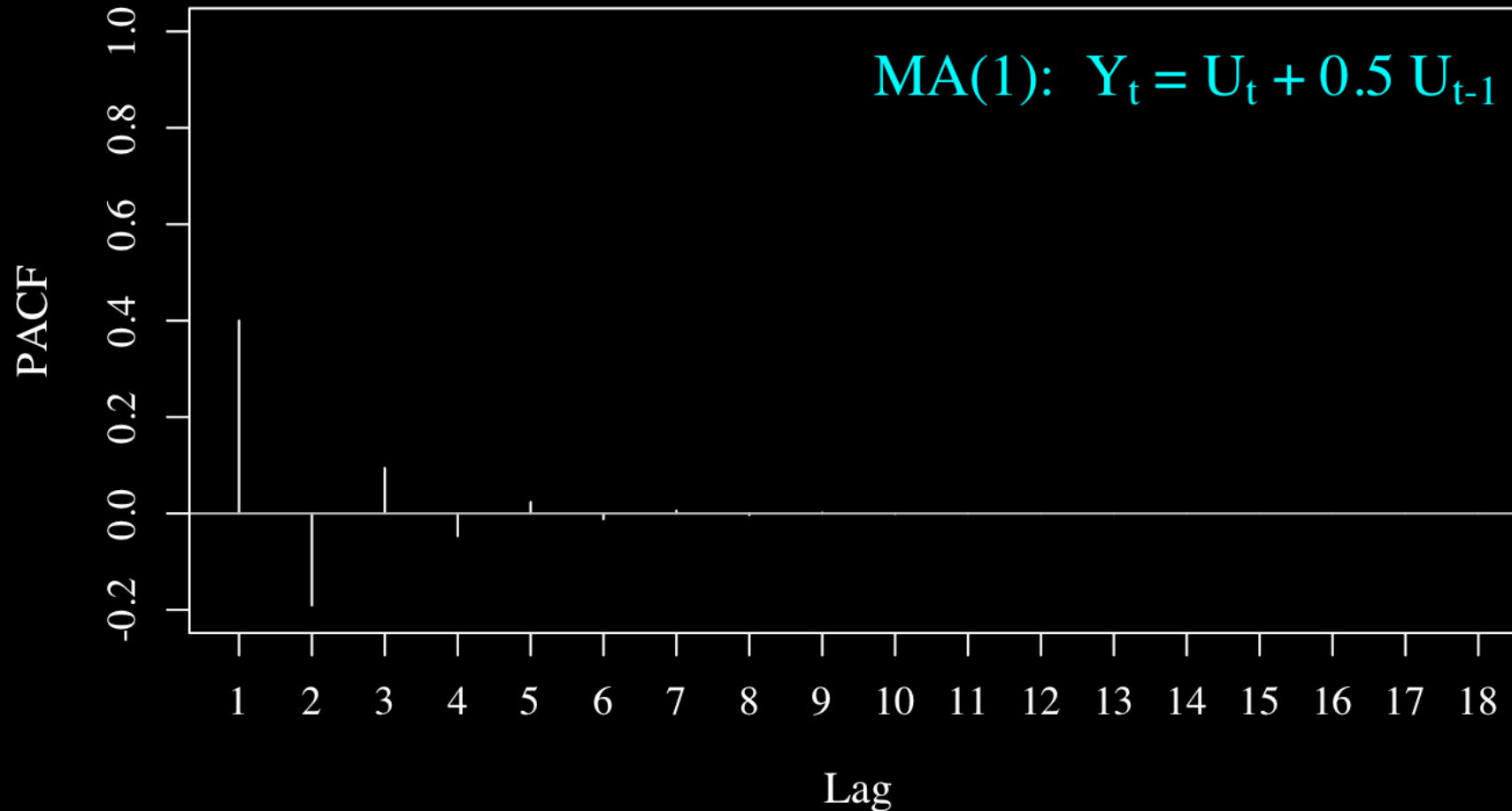
Theoretical partial autocorrelation function

```
1 pacf_AR1 <- ARMAacf(ar=0.4, pacf=TRUE, la
```



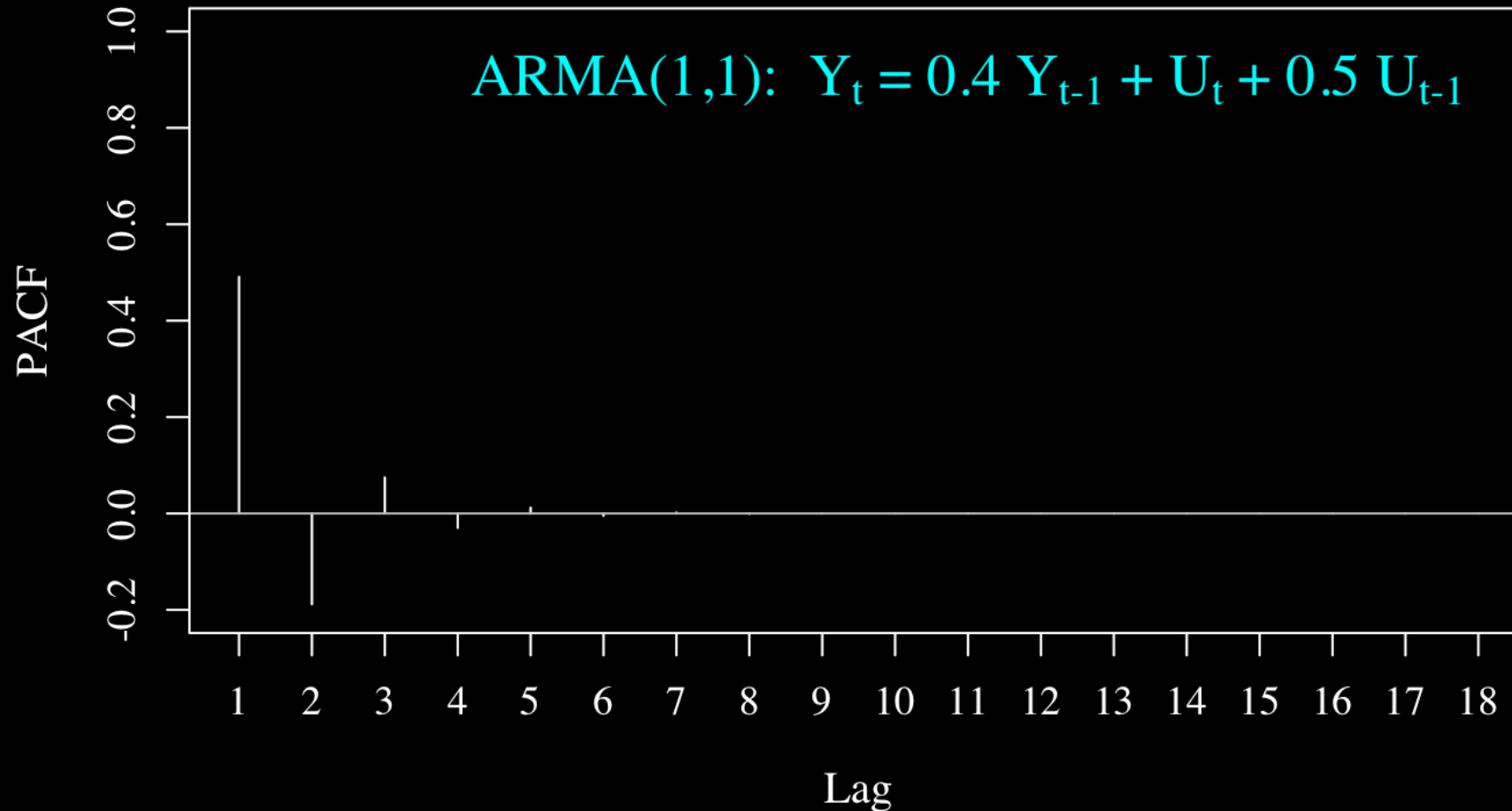
Theoretical partial autocorrelation function

```
1 pacf_MA1 <- ARMAacf(ma=0.5, pacf=TRUE, la
```



Theoretical partial autocorrelation function

```
1 pacf_ARMA11 <- ARMAacf(ar=0.2, ma=0.4, pacf=)
```



Theoretical ACF and PACF patterns

As lag increases:

	ACF	PACF
$AR(p)$	declines	zero after p lags
$MA(q)$	zero after q lags	declines
$ARMA(p, q)$	declines	declines

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- *Assume* $\text{var}(U_t) = \sigma^2$ for every t .

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\text{cov}(Z_t, Z_{t-1}) = \text{cov}(U_t + \theta_1 U_{t-1}, U_{t-1} + \theta_1 U_{t-2})$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\begin{aligned} \text{cov}(Z_t, Z_{t-1}) &= \text{cov}(U_t + \theta_1 U_{t-1}, U_{t-1} + \theta_1 U_{t-2}) \\ &= \text{cov}(U_t, U_{t-1}) + \theta_1 \text{cov}(U_{t-1}, U_{t-1}) \\ &\quad + \theta_1 \text{cov}(U_t, U_{t-2}) \\ &\quad + \theta_1^2 \text{cov}(U_{t-1}, U_{t-2}) \end{aligned}$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\begin{aligned} \text{cov}(Z_t, Z_{t-1}) &= \text{cov}(U_t, U_{t-1}) + \theta_1 \text{cov}(U_{t-1}, U_{t-1}) \\ &\quad + \theta_1 \text{cov}(U_t, U_{t-2}) \\ &\quad + \theta_1^2 \text{cov}(U_{t-1}, U_{t-2}) \end{aligned}$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\begin{aligned} \text{cov}(Z_t, Z_{t-1}) &= \text{cov}(U_t, U_{t-1}) + \theta_1 \text{cov}(U_{t-1}, U_{t-1}) \\ &\quad + \theta_1 \text{cov}(U_t, U_{t-2}) \\ &\quad + \theta_1^2 \text{cov}(U_{t-1}, U_{t-2}) \end{aligned}$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\begin{aligned} \text{cov}(Z_t, Z_{t-1}) = & \quad 0 \quad + \theta_1 \sigma^2 \\ & + \theta_1 \quad 0 \\ & + \theta_1^2 \quad 0 \end{aligned}$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\text{cov}(Z_t, Z_{t-1}) = \theta_1 \sigma^2$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

$$\begin{aligned} \text{cov}(Z_t, Z_{t-2}) &= \text{cov}(U_t, U_{t-2}) + \theta_1 \text{cov}(U_{t-1}, U_{t-2}) \\ &\quad + \theta_1 \text{cov}(U_t, U_{t-3}) \\ &\quad + \theta_1^2 \text{cov}(U_{t-1}, U_{t-3}) \\ &= 0 \end{aligned}$$

Example: MA(1)

$$Z_t = U_t + \theta_1 U_{t-1}, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

- Recall $\text{cov}(U_t, U_{t-k}) = 0$ for every $k = 1, 2, \dots$
- Assume $\text{var}(U_t) = \sigma^2$ for every t .

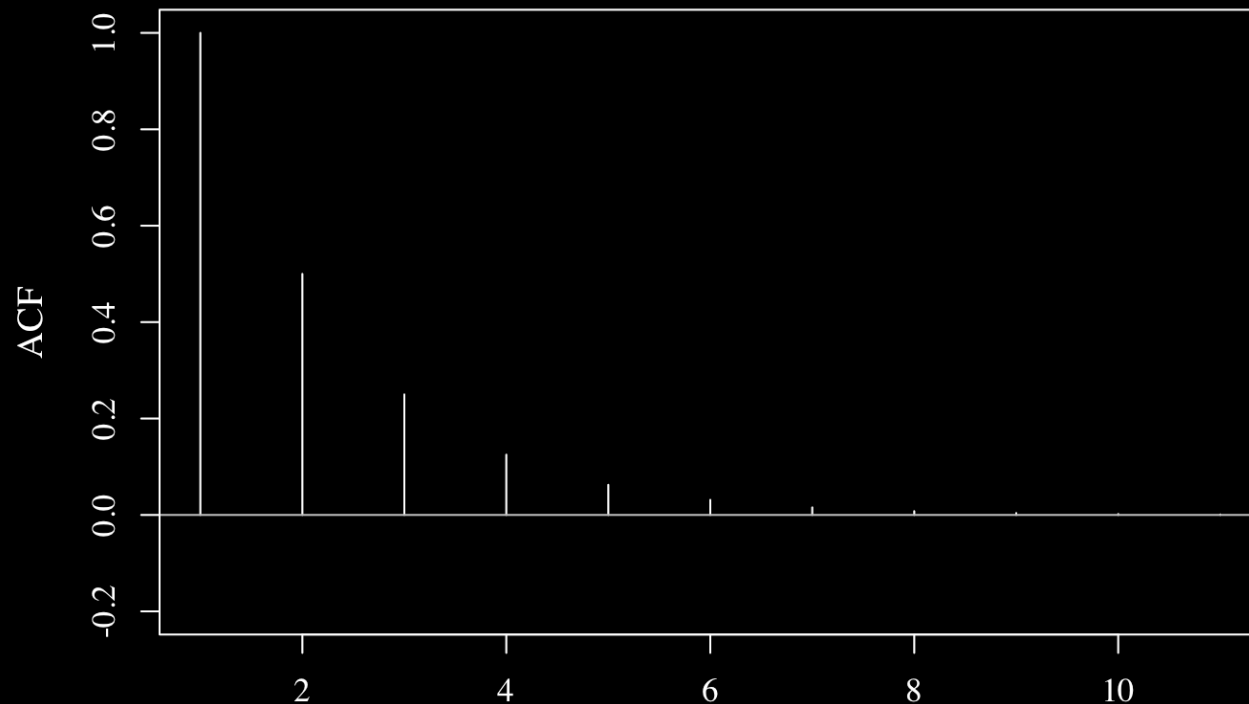
For any $k > 1$:

$$\begin{aligned} \text{cov}(Z_t, Z_{t-k}) &= \text{cov}(U_t, U_{t-k}) + \theta_1 \text{cov}(U_{t-1}, U_{t-k}) \\ &\quad + \theta_1 \text{cov}(U_t, U_{t-k-1}) \\ &\quad + \theta_1^2 \text{cov}(U_{t-1}, U_{t-k-1}) \\ &= 0 \end{aligned}$$

Example: AR(1)

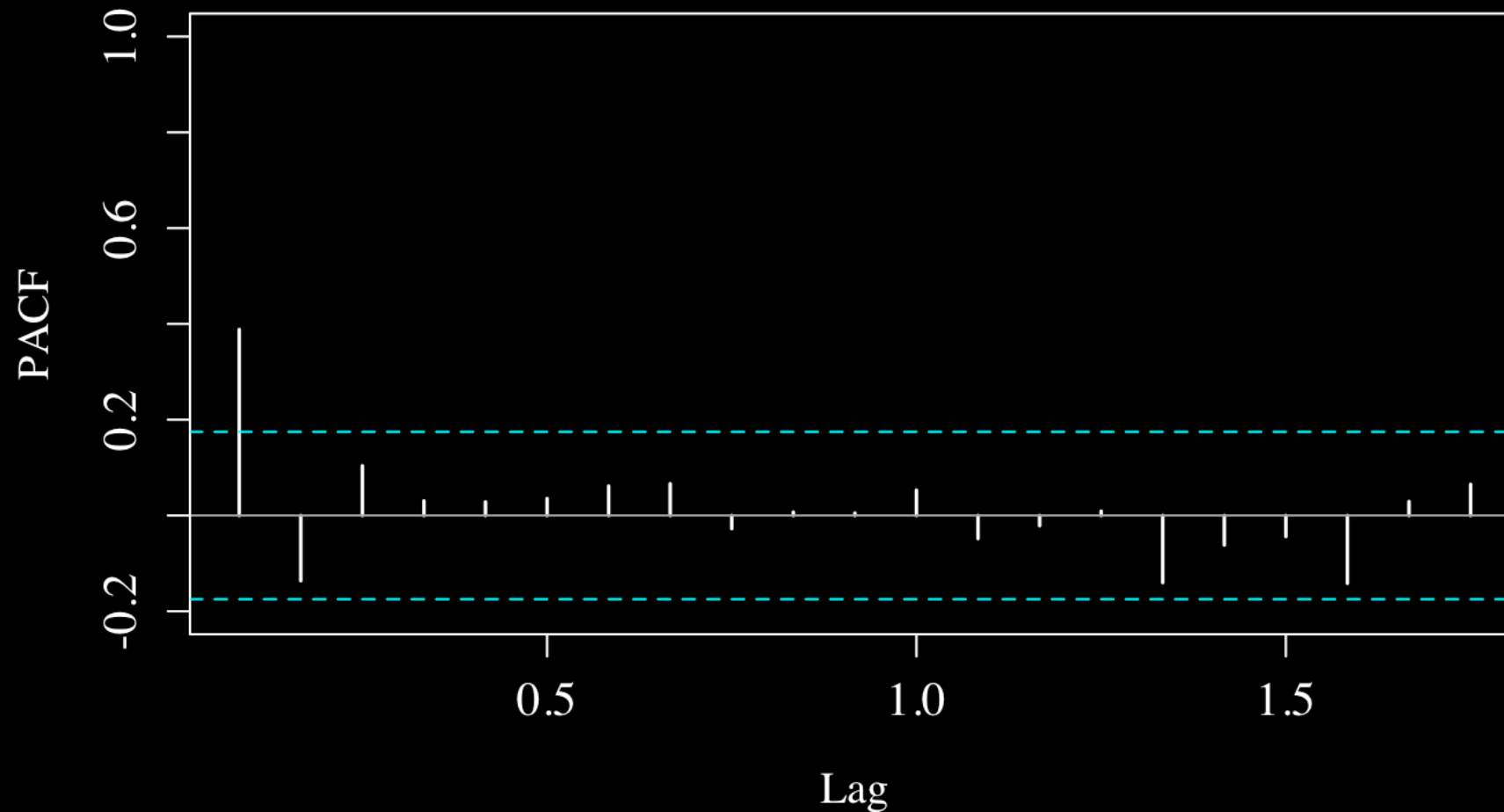
$$Z_t = \phi_1 Z_{t-1} + U_t, \quad U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

If $|\phi_1| < 1$ then $\text{cov}(Z_t, Z_{t-k}) = \phi_1^k$. (*Details later*)



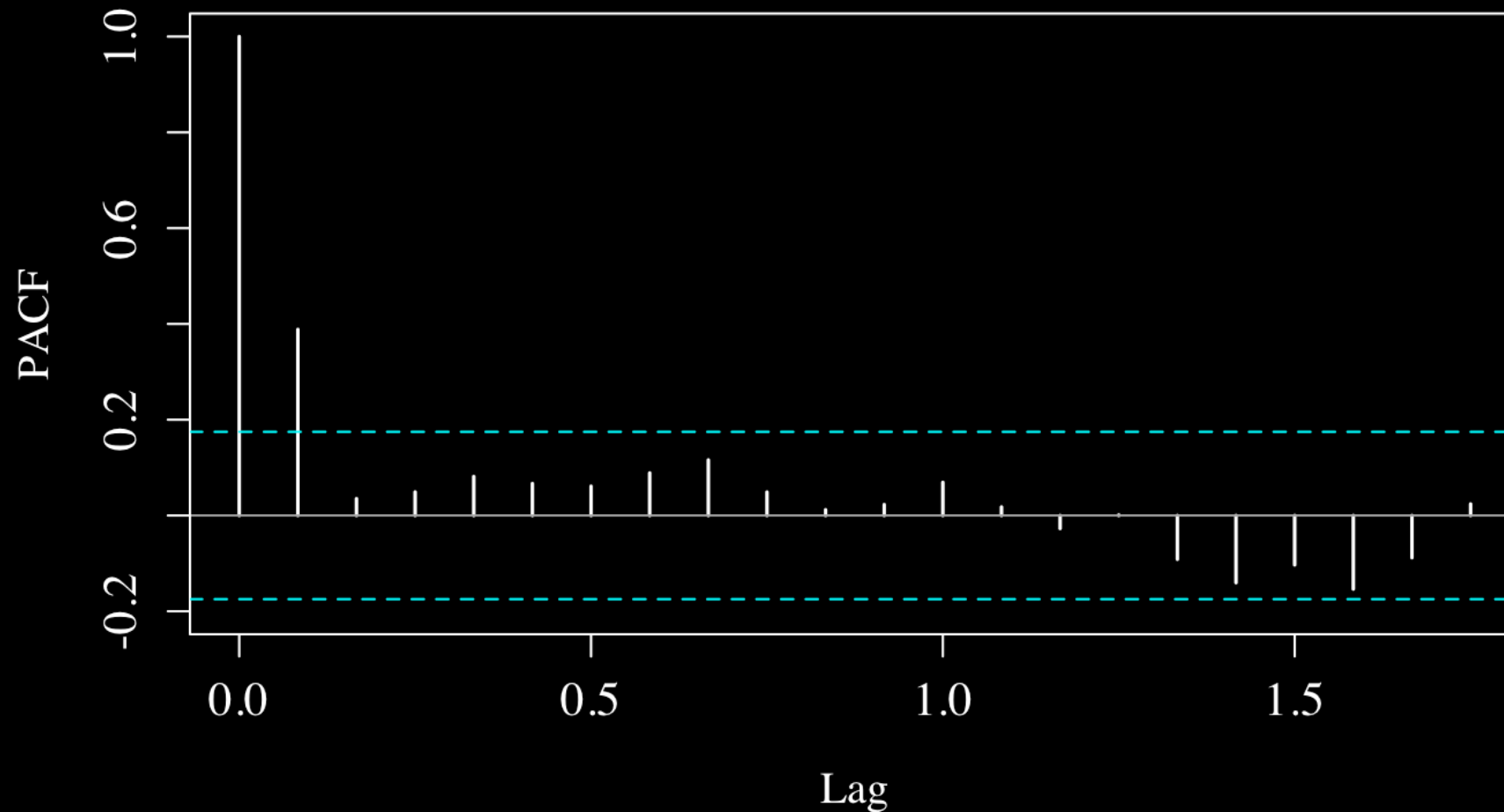
First difference of interest rates

```
1 acf(DY, type="partial")
```



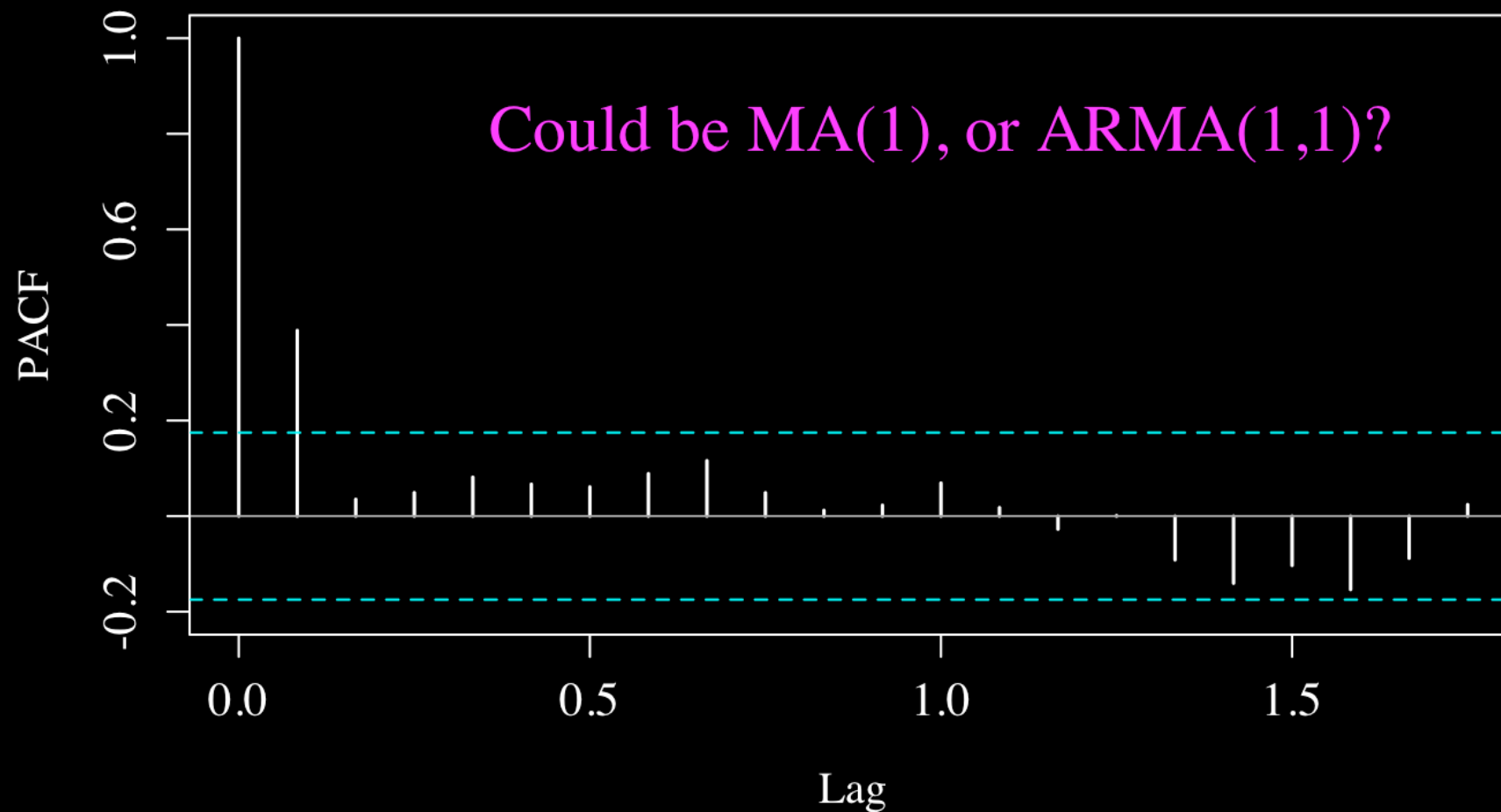
First difference of interest rates

1 `acf(DY)`



First difference of interest rates

```
1 acf(DY)
```



Model estimation: AR(1)

```
1 AR1 <- Arima(DY, order=c(1,0,0))  
2 print(AR1)
```

Series: DY

ARIMA(1,0,0) with non-zero mean

Coefficients:

	ar1	mean
	0.3858	-0.0318
s.e.	0.0801	0.0145

Model estimation: MA(1)

```
1 MA1 <- Arima(DY, order=c(0,0,1))  
2 print(MA1)
```

Series: DY

ARIMA(0,0,1) with non-zero mean

Coefficients:

	ma1	mean
	0.4434	-0.0320
s.e.	0.0762	0.0127

Model estimation: ARMA(1,1)

```
1 ARMA11 <- Arima(DY, order=c(1,0,1))  
2 print(ARMA11)
```

Series: DY

ARIMA(1,0,1) with non-zero mean

Coefficients:

	ar1	ma1	mean
	0.0558	0.3989	-0.032
s.e.	0.1915	0.1758	0.013

Residual autocorrelation testing

```
1 LBp_AR1 <- checkresiduals(AR1)
```

Ljung-Box test

data: Residuals from ARIMA(1,0,0) with non-zero mean

$Q^* = 19$, $df = 23$, $p\text{-value} = 0.7012$ ✓

Model df: 1. Total lags used: 24

Residual autocorrelation testing

```
1 LBp_MA1 <- checkresiduals(MA1)
```

Ljung-Box test

data: Residuals from ARIMA(0,0,1) with non-zero mean

$Q^* = 16.201$, $df = 23$, $p\text{-value} = 0.8465$ ✓

Model df: 1. Total lags used: 24

Residual autocorrelation testing

```
1 LBP_ARMA11 <- checkresiduals(ARMA11)
```

Ljung-Box test

data: Residuals from ARIMA(1,0,1) with non-zero mean

$Q^* = 15.748$, $df = 22$, $p\text{-value} = 0.8282$ ✓

Model df: 2. Total lags used: 24

Model selection: AICc

	AICc
AR1	-219.24
MA1	-222.58 ✓
ARMA11	-220.54

- All three models pass autocorrelation tests.
- The AICc prefers the MA(1) model.

Summary

One step ahead prediction error:

$$U_t = Z_t - E(Z_t | \mathcal{Z}_{t-1})$$

$$\text{MA}(q) : \quad Z_t = U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q}$$

$$\begin{aligned} \text{ARMA}(p, q) : \quad Z_t = & \phi_1 Z_{t-1} + \dots + \phi_p Z_{t-p} \\ & + U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q} \end{aligned}$$

Select p, q using AICc and autocorrelation tests.

Summary

As lag increases:

	ACF	PACF
$AR(p)$	declines	zero after p lags
$MA(q)$	zero after q lags	declines
$ARMA(p, q)$	declines	declines